FUNCTIONAL INEQUALITIES AND SYNCHRONIZATION FOR THE MEASURE-VALUED SOLUTIONS OF STOCHASTIC DIFFERENTIAL EQUATIONS WITH INTERACTION

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Abstract. We study the measure-valued solution to the stochastic differential equations with interaction. We derive pathwise transfer estimates for logarithmic Sobolev, Poincaré, concentration, and T_2 inequalities. The resulting constants are controlled by the maximal Jacobian norm or the Lipschitz distortion of the flow on the support of the initial measure. Under global spatial regularity assumptions, we obtain moment estimates for the maximal Jacobian norm on compact sets. We also establish an averaged Jacobian-energy bound under one-sided derivative assumptions. Finally, for a kernel-type interacting model satisfying a pairwise dissipativity condition, we prove exponential decay of the expected pairwise variance, yielding mean-square synchronization around the random center of mass.

Contents

1. Introduction

We consider stochastic differential equations with interaction that were proposed by Andrey A. Dorogovtsev in [Dorogovtsev, 2003],

$$\begin{cases} dX_t(u) &= a\big(X_t(u),\mu_t\big)\,dt + \displaystyle\sum_{\ell=1}^{r}\sigma_\ell\big(X_t(u)\big)\,dB_t^\ell, \\ X_0(u) &= u, \\ \mu_t &= \mu_0\circ X_t^{-1}. \end{cases} \tag{1}$$

Here μ_0 is a deterministic initial probability measure, where $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of probability measures on $\mathbb{R}^d$ with finite second moment. $a:\mathbb{R}^d\times\mathcal{P}_2(\mathbb{R}^d)\to\mathbb{R}^d$ is the drift and the diffusion coefficient $\sigma_\ell:\mathbb{R}^d\to\mathbb{R}^d$, $\ell=1,\dots,r$, is state-dependent and

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non-constant. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions and carrying a r-dimensional Brownian motion $B = (B^1, \ldots, B^r)$.

We write $|\cdot|$ for the Euclidean norm on $\mathbb{R}^d$. For a matrix $M \in \mathbb{R}^{d\times d}$ we set

$$\|M\|_{\mathrm{HS}} := \Big(\sum_{i,j=1}^{d} |M_{ij}|^2 \Big)^{1/2}.$$

Fix $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$.

Assumption 1 (Well-posedness)**.** *There exists $L < \infty$ such that, for all $x, y \in \mathbb{R}^d$ and $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$|a(x,\mu) - a(y,\nu)| + \left(\sum_{\ell=1}^{r} |\sigma_\ell(x) - \sigma_\ell(y)|^2 \right)^{1/2} \leq L\big(|x-y| + W_2(\mu,\nu)\big)$$

Under Assumption 1, the SDEWI (1) admits a unique global solution; see [Dorogovtsev, 2023, Theorem 2.3.1]. This assumption remains in force throughout Sections 2 and 3. Additional spatial regularity conditions are stated locally when they are needed. Section 4 concerns a different kernel model and is treated under separate assumptions.

Unlike a classical stochastic differential equation, the drift depends on the push-forward of the entire system through the random measure μ_t. This dependence represents the interaction between particles. For each realization $\omega \in \Omega$, the map

$$T_t(\omega, \cdot) : u \mapsto X_t(u, \omega)$$

acts on the initial measure by pushforward,

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1},$$

so the evolution is best viewed as a random transport of measures by a stochastic flow (1). The central question of this paper is how the random transport T_t deforms the properties of the initial measure μ_0. Whenever the flow is differentiable with respect to the initial label, we write

$$J_t(u) := D_u T_t(u),$$

and consider the three quantities described below.

Firstly, we consider the maximal Jacobian norm $\mathcal{K}_t(K)$ measures the worst-case infinitesimal stretching of the transport on the compact set K,

$$\mathcal{K}_t(K) := \sup_{u\in K} \|J_t(u)\|_{\mathrm{op}},$$

The Jacobian energy $\mathcal{E}_t$

$$\mathcal{E}_t := \int_{\mathbb{R}^d} \|J_t(u)\|_{\mathrm{HS}}^2 \,\mu_0(du),$$

measures the same infinitesimal deformation after averaging over the initial mass.

The pairwise energy $\mathcal{D}_t$

$$\mathcal{D}_t := \iint_{\mathbb{R}^d\times\mathbb{R}^d} |T_t(u) - T_t(v)|^2 \,\mu_0(du)\mu_0(dv).$$

measures deformation at finite scale by averaging the squared distance between two transported labels.

These quantities describe different spatial resolutions of the same random transport. The first is local and unform in the initial label, the second is local but averaged over the initial distribution, and the third is averaged and finite-scale. Correspondingly, they lead to different conclusions and require different assumptions. Uniform Jacobian control is strong enough to transfer functional inequalities pathwise, but it requires substantial spatial regularity. Averaged Jacobian energy can be estimated under weaker one-sided derivative conditions. Pairwise energy can be controlled directly through dissipativity of the dynamics, without first obtaining a uniform Jacobian estimate.

Our results follow these three levels of deformation. In Section 2, we study the strongest, worst-case infinitesimal quantity $\mathcal{K}_t(K)$. We prove moment estimates for this random distortion constant on compact sets and show that it governs the pathwise transfer of logarithmic Sobolev, Poincaré, concentration, and transport inequalities from μ_0 to μ_t.

In Section 3, we lower the level of spatial resolution and replace the supremum over labels by the averaged Jacobian energy $\mathcal{E}_t$. This quantity does not provide the same uniform pathwise information, but it admits a direct estimate under weaker one-sided assumptions on the derivative coefficients.

Finally, in Section 4, we pass from infinitesimal deformation to finite-scale deformation. For a kernel-type interacting model, we impose a pairwise dissipativity condition and prove exponential decay of $\mathbb{E}\mathcal{D}_t$. This decay means that the transported labels approach one another in mean square and that the random measure synchronizes around its random center of mass.

2. Pathwise Jacobian Bounds and Functional Inequality Transfer

This section has two purposes. First, we quantify the spatial distortion of the stochastic flow T_t through its Jacobian J_t. Second, we use this distortion to transfer functional inequalities from the initial measure μ_0 to the random pushforward measure μ_t. The technical proof of the compact-set Jacobian estimate is postponed to the final subsection.

Assumption 2 (Spatial differentiability)**.** *For every $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, the map $x \mapsto a(x,\mu)$ belongs to $C^1(\mathbb{R}^d;\mathbb{R}^d)$, and*

$$\sigma_\ell \in C^1(\mathbb{R}^d;\mathbb{R}^d), \qquad \ell = 1,\dots,r.$$

By Assumptions 1 and 2,

$$\|D_x a(x,\mu)\|_{\mathrm{op}} \le L_x, \qquad \sum_{\ell=1}^{r} |D\sigma_\ell(x)z|^2 \le L_\sigma^2 |z|^2 \tag{2}$$

for all $x, z \in \mathbb{R}^d$ and $\mu \in \mathcal{P}_2(\mathbb{R}^d)$

2.1. **The Jacobian flow and distortion constants.**

We first identify the Jacobian equation for the stochastic flow and introduce the distortion quantities used throughout the section.

For a fixed ω and t, define the flow map $T_t(\omega,u) := X_t(u,\omega)$ and its Jacobian matrix

$$J_t(u) := \partial_u T_t(u) = \partial_u X_t(u) \in \mathbb{R}^{d\times d}.$$

We use the Jacobian SDE from [Dorogovtsev and Weiß, 2023]. For equation (1), it has the following form.

Theorem 2.1 (Jacobian equation)**.** *Assume Assumptions 1 and 2. Then the solution admits a modification such that, for almost every ω and every $t \ge 0$, the map*

$$u \longmapsto T_t(\omega,u)$$

is of class C^1. Its Jacobian $J_t(u)$ satisfies

$$\begin{cases} dJ_t(u) = A_t(u)J_t(u)\,dt + \sum_{\ell=1}^{r} C_{\ell,t}(u)J_t(u)\,dB_t^\ell, \\ J_0(u) = I, \end{cases} \tag{3}$$

where

$$A_t(u) := D_x a\big(X_t(u),\mu_t\big), \qquad C_{\ell,t}(u) := D\sigma_\ell\big(X_t(u)\big).$$

In particular,

$$\|A_t(u)\|_{\mathrm{op}} \le L_x, \qquad \sum_{\ell=1}^{r} |C_{\ell,t}(u)z|^2 \le L_\sigma^2|z|^2$$

for all t,u,z.

Definition 2.1 (Random Jacobian constant)**.** For a nonempty set $K \subset \mathbb{R}^d$, define

$$\mathcal{K}_t(\omega;K) := \sup_{u\in K} \|J_t(u,\omega)\|_{\mathrm{op}}.$$

If K is compact and $u \mapsto J_t(u,\omega)$ is continuous, then $\mathcal{K}_t(\omega;K) < \infty$.

If K is compact, continuity of $u \mapsto J_t(u)$ implies $\mathcal{K}_t(K) < \infty$. This compact-set quantity is used because a corresponding supremum over all of $\mathbb{R}^d$ need not be finite. The counterexamples of Mohammed–Scheutzow show that even for globally Lipschitz coefficients one may fail to have a finite global constant $\sup_{u\in\mathbb{R}^d} \|J_t(u,\omega)\|_{\mathrm{op}}$ for a fixed (t,ω).

Lemma 2.1 (Flow Lipschitz constant via Jacobian matrix)**.** *For each fixed (t,ω) such that the flow map $u \mapsto T_t(\omega,u) := X_t(u,\omega)$ is of class C^1,*

$$\mathrm{Lip}(T_t(\omega)) = \sup_{u\in\mathbb{R}^d} \|J_t(u,\omega)\|_{\mathrm{op}} \in [0,\infty].$$

Proof. Since $u \mapsto T_t(\omega, u)$ is C^1, the mean value theorem implies

$$|T_t(u) - T_t(v)| \le \sup_{\theta \in [0,1]} \|J_t(v + \theta(u - v))\|_{\mathrm{op}} \, |u - v|.$$

Taking the supremum over $u \neq v$ yields $\mathrm{Lip}(T_t) \le \sup_u \|J_t(u)\|_{\mathrm{op}}$. Conversely, for any u and unit vector ξ, letting $v = u + \varepsilon\xi$ and sending $\varepsilon \downarrow 0$ gives

$$\|J_t(u)\|_{\mathrm{op}} = \sup_{|\xi|=1} \lim_{\varepsilon \downarrow 0} \frac{|T_t(u + \varepsilon\xi) - T_t(u)|}{\varepsilon} \le \mathrm{Lip}(T_t),$$

hence equality. □

Pointwise estimates for $J_t(u)$ follow from the boundedness of the coefficients in the linear equation (3). Controlling the compact-set maximal Jacobian norm $\mathcal{K}_t(K)$, however, requires estimates for the spatial increments $J_t(u) - J_t(v)$. The additional regularity needed for this purpose is therefore imposed only in the following theorem.

Theorem 2.2 (Moments of the compact-set maximal Jacobian norm)**.** *Assume Assumptions 1 and 2. Suppose in addition that there exists a constant $H < \infty$ such that, for all $x, y, z \in \mathbb{R}^d$ and $\mu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\big|\big(D_x a(x,\mu) - D_x a(y,\mu)\big)z\big| + \left(\sum_{\ell=1}^{r} \big|\big(D\sigma_\ell(x) - D\sigma_\ell(y)\big)z\big|^2\right)^{1/2} \le H|x-y|\,|z|. \tag{4}$$

Then, for every nonempty compact $K \subset \mathbb{R}^d$ and every $p \ge 1$, there exist constants $C_{p,K}, c_p < \infty$ such that

$$\mathbb{E}\big[\mathcal{K}_t(\cdot; K)^p\big] \le C_{p,K} e^{c_p t}, \qquad t \ge 0. \tag{5}$$

The proof is given in Section 2.3.

2.2. **Transfer of functional inequalities.** In this subsection we record several consequences of the identity

$$T_t(\omega, u) := X_t(u, \omega),$$

which expresses the random measure μ_t as the pushforward of the initial law μ_0 under the stochastic flow.

Definition 2.2 (Logarithmic Sobolev inequality)**.** Denoting by $\mathcal{C}_c^1(\mathbb{R}^d)$ the set of continuously differentiable compactly supported functions from $\mathbb{R}^d$ to $\mathbb{R}$, for $g \ge 0$ with $\int g \, d\mu = 1$ we define the entropy

$$\mathrm{Ent}_\mu(g) := \int g \log g \, d\mu.$$

A probability measure μ on $\mathbb{R}^d$ is said to satisfy a logarithmic Sobolev inequality with constant $C > 0$ if for every $h \in \mathcal{C}_c^1(\mathbb{R}^d)$ with $\int h^2 \, d\mu = 1$ one has

$$\mathrm{Ent}_\mu(h^2) \le C \int_{\mathbb{R}^d} |\nabla h|^2 \, d\mu. \tag{6}$$

Corollary 2.1 (Pathwise LSI with the Jacobian matrix constant)**.** *If μ_0 satisfies* $\mathrm{LSI}(C_0)$ *and* $\mathrm{supp}(\mu_0) \subset K$ *for some compact K, then for every $t \ge 0$ and a.s. ω,*

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1} \quad \Longrightarrow \quad \mu_t(\omega) \text{ satisfies } \mathrm{LSI}\big(C_0\, \mathcal{K}_t(\omega; K)^2\big).$$

We next record the basic concentration-transfer principle under a random Lipschitz map.

Proposition 2.1 (Pathwise concentration transfer)**.** *Let $K := \mathrm{supp}(\mu_0)$ and assume that for a.s. ω the map $T_t(\omega, \cdot)$ is Lipschitz on K, with Lipschitz constant*

$$L_t(\omega; K) := \sup_{\substack{u,v \in K \\ u \neq v}} \frac{|T_t(\omega, u) - T_t(\omega, v)|}{|u - v|}.$$

Assume that μ_0 satisfies the concentration inequality

$$\mu_0(f - \mu_0(f) \geq r) \leq \Psi(r), \qquad r \geq 0, \tag{7}$$

for every 1*-Lipschitz function $f : \mathbb{R}^d \to \mathbb{R}$, where $\Psi : [0, \infty) \to [0, 1]$ is nonincreasing. Then for a.s. ω,*

$$\mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) \leq \Psi\Big(\frac{r}{L_t(\omega; K)}\Big), \qquad r \geq 0, \tag{8}$$

for every 1*-Lipschitz function $g : \mathbb{R}^d \to \mathbb{R}$.*

Proof. Fix ω such that $T_t(\omega, \cdot)$ is Lipschitz on K, and write $L := L_t(\omega; K) \in [0, \infty)$. Let $g : \mathbb{R}^d \to \mathbb{R}$ be 1-Lipschitz.

If $L = 0$, then $T_t(\omega, \cdot)$ is constant on K, hence $\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1}$ is a Dirac mass and $\mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) = 0$ for every $r > 0$. Thus equation (8) is immediate.

Assume now $L > 0$ and define on K

$$h(u) := \frac{g(T_t(\omega, u))}{L}.$$

Then h is 1-Lipschitz on K. By the McShane extension theorem, h extends to a 1-Lipschitz function $\widetilde{h} : \mathbb{R}^d \to \mathbb{R}$. Applying equation (7) to $\widetilde{h}$ (and using that μ_0 is supported on K) gives

$$\mu_0\big(h - \mu_0(h) \geq s\big) \leq \Psi(s), \qquad s \geq 0.$$

Since $\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1}$,

$$\mu_t(\omega)(g) = \mu_0(g \circ T_t(\omega, \cdot)) = L\mu_0(h).$$

Therefore, for $r \geq 0$,

$$\begin{aligned}
\mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) &= \mu_0\big(g \circ T_t(\omega, \cdot) - \mu_0(g \circ T_t(\omega, \cdot)) \geq r\big) \\
&= \mu_0\big(h - \mu_0(h) \geq r/L\big) \\
&\leq \Psi\Big(\frac{r}{L}\Big) = \Psi\Big(\frac{r}{L_t(\omega; K)}\Big).
\end{aligned}$$

This proves equation (8). $\square$

Remark 2.1 (Concentration as a corollary of LSI). If μ_0 satisfies a logarithmic Sobolev inequality, then it satisfies Gaussian concentration. Hence Corollary 2.1 yields, in particular, a pathwise concentration inequality for $\mu_t(\omega)$ with a random constant controlled by $\mathcal{K}_t(\omega; K)^2$.

Definition 2.3 (Poincaré inequality)**.** A probability measure μ on $\mathbb{R}^d$ is said to satisfy a Poincaré inequality with constant $C_P > 0$, denoted by $\mathrm{PI}(C_P)$, if for every $h \in \mathcal{C}_c^1(\mathbb{R}^d)$,

$$\mathrm{Var}_\mu(h) := \int_{\mathbb{R}^d} h^2 \, d\mu - \Big(\int_{\mathbb{R}^d} h \, d\mu\Big)^2 \leq C_P \int_{\mathbb{R}^d} |\nabla h|^2 \, d\mu. \tag{9}$$

We first record the stability of the Poincaré inequality under Lipschitz pushforward.

Proposition 2.2 (Poincaré inequality under a Lipschitz pushforward)**.** *Let μ be a probability measure on $\mathbb{R}^d$ satisfying $\mathrm{PI}(C_P)$, and let $T : \mathbb{R}^d \to \mathbb{R}^d$ be a Lipschitz map with Lipschitz constant $\mathrm{Lip}(T)$. Then the pushforward measure $\nu := \mu \circ T^{-1}$ satisfies*

$$\mathrm{PI}\big(C_P \, \mathrm{Lip}(T)^2\big).$$

Proof. Let $g \in \mathcal{C}_c^1(\mathbb{R}^d)$ and set $f := g \circ T$. Then

$$\int_{\mathbb{R}^d} g \, d\nu = \int_{\mathbb{R}^d} g(T(u)) \, \mu(du) = \int_{\mathbb{R}^d} f(u) \, \mu(du),$$

and similarly

$$\int_{\mathbb{R}^d} g^2 \, d\nu = \int_{\mathbb{R}^d} f^2 \, d\mu.$$

Hence

$$\mathrm{Var}_\nu(g) = \mathrm{Var}_\mu(f).$$

Applying the Poincaré inequality for μ gives

$$\mathrm{Var}_\nu(g) = \mathrm{Var}_\mu(f) \le C_P \int_{\mathbb{R}^d} |\nabla f(u)|^2 \, \mu(du).$$

Since $f = g \circ T$ and T is Lipschitz, we have a.e.

$$|\nabla f(u)| \le \mathrm{Lip}(T) \, |\nabla g(T(u))|.$$

Therefore

$$\mathrm{Var}_\nu(g) \le C_P \ \mathrm{Lip}(T)^2 \int_{\mathbb{R}^d} |\nabla g(T(u))|^2 \, \mu(du) = C_P \ \mathrm{Lip}(T)^2 \int_{\mathbb{R}^d} |\nabla g(x)|^2 \, \nu(dx).$$

This proves the claim. □

Corollary 2.2 (Pathwise Poincaré inequality). *If μ_0 satisfies $\mathrm{PI}(C_P)$ and $\mathrm{supp}(\mu_0) \subset K$ for some compact set $K \subset \mathbb{R}^d$, then for every $t \ge 0$ and a.s. ω,*

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1} \quad \Longrightarrow \quad \mu_t(\omega) \ \text{satisfies} \ \mathrm{PI}\big(C_P \, \mathcal{K}_t(\omega; K)^2\big).$$

Proof. By Proposition 2.2, it suffices to bound the Lipschitz constant of $T_t(\omega, \cdot)$ on K. Since K is compact and $u \mapsto T_t(\omega, u)$ is of class C^1, the mean value theorem gives

$$|T_t(\omega, u) - T_t(\omega, v)| \le \sup_{z \in K} \|J_t(z, \omega)\|_{\mathrm{op}} \, |u - v| = \mathcal{K}_t(\omega; K) \, |u - v|, \qquad u, v \in K.$$

Hence $\mathrm{Lip}(T_t(\omega)|_K) \le \mathcal{K}_t(\omega; K)$, and the result follows. □

Definition 2.4 (Talagrand transport inequality). Let $p \ge 1$. A probability measure μ on $\mathbb{R}^d$ is said to satisfy the Talagrand transport inequality $T_p(C_T)$ with constant $C_T > 0$ if for every probability measure ν on $\mathbb{R}^d$ with finite p-th moment,

$$W_p(\nu, \mu)^2 \le 2 C_T \ \mathrm{Ent}(\nu \mid \mu), \tag{10}$$

where

$$\mathrm{Ent}(\nu \mid \mu) = \begin{cases} \displaystyle\int_{\mathbb{R}^d} \log\left(\frac{d\nu}{d\mu}\right) d\nu, & \nu \ll \mu, \\ +\infty, & \text{otherwise.} \end{cases}$$

We next record the analogous stability property for Talagrand transport inequalities.

Proposition 2.3 (Transport inequality under a Lipschitz pushforward). *Let μ be a probability measure on $\mathbb{R}^d$ satisfying $T_p(C_T)$ for some $p \ge 1$, and let $T : \mathbb{R}^d \to \mathbb{R}^d$ be a Lipschitz map with Lipschitz constant $\mathrm{Lip}(T)$. Then the pushforward measure $\eta := \mu \circ T^{-1}$ satisfies*

$$T_p\big(C_T \ \mathrm{Lip}(T)^2\big).$$

Proof. Let λ be any probability measure on $\mathbb{R}^d$. If $\lambda \not\ll \eta$, then $\mathrm{Ent}(\lambda \mid \eta) = +\infty$ and the inequality is trivial. Assume therefore that $\lambda \ll \eta$.

Choose a probability measure $\widetilde{\lambda}$ on $\mathbb{R}^d$ such that

$$\widetilde{\lambda} \circ T^{-1} = \lambda \qquad \text{and} \qquad \mathrm{Ent}(\widetilde{\lambda} \mid \mu) = \mathrm{Ent}(\lambda \mid \eta).$$

For instance, one may take the measure

$$\widetilde{\lambda}(du) := \frac{d\lambda}{d\eta}(T(u)) \, \mu(du).$$

Indeed,

$$(\widetilde{\lambda} \circ T^{-1})(A) = \int_{\mathbb{R}^d} \mathbf{1}_{\{T(u) \in A\}} \frac{d\lambda}{d\eta}(T(u)) \, \mu(du) = \int_A \frac{d\lambda}{d\eta}(x) \, \eta(dx) = \lambda(A),$$

and

$$\mathrm{Ent}(\widetilde{\lambda} \mid \mu) = \int_{\mathbb{R}^d} \log\Big(\frac{d\lambda}{d\eta}(T(u))\Big)\,\widetilde{\lambda}(du) = \int_{\mathbb{R}^d} \log\Big(\frac{d\lambda}{d\eta}(x)\Big)\,\lambda(dx) = \mathrm{Ent}(\lambda \mid \eta).$$

Now let π be a coupling of $\widetilde{\lambda}$ and μ. Then $\pi \circ (T,T)^{-1}$ is a coupling of λ and η, so by Lipschitz continuity of T,

$$\begin{aligned}
&\int_{\mathbb{R}^d\times\mathbb{R}^d} |x-y|^p\, d(\pi\circ(T,T)^{-1})(x,y)\\
&\quad = \int_{\mathbb{R}^d\times\mathbb{R}^d} |T(u)-T(v)|^p\, d\pi(u,v)\\
&\quad \le \mathrm{Lip}(T)^p \int |u-v|^p\, d\pi(u,v).
\end{aligned}$$

Taking the infimum over all couplings π of $\widetilde{\lambda}$ and μ yields

$$W_p(\lambda,\eta) \le \mathrm{Lip}(T)\, W_p(\widetilde{\lambda},\mu).$$

Since μ satisfies $T_p(C_T)$,

$$W_p(\widetilde{\lambda},\mu)^2 \le 2C_T\ \mathrm{Ent}(\widetilde{\lambda} \mid \mu) = 2C_T\ \mathrm{Ent}(\lambda \mid \eta).$$

Combining the two inequalities, we obtain

$$W_p(\lambda,\eta)^2 \le \mathrm{Lip}(T)^2\, W_p(\widetilde{\lambda},\mu)^2 \le 2C_T\ \mathrm{Lip}(T)^2\ \mathrm{Ent}(\lambda \mid \eta),$$

which proves that η satisfies $T_p(C_T\ \mathrm{Lip}(T)^2)$. □

2.3. **Proof of the compact-set Jacobian estimate.** The proof of Theorem 2.2 relies on two elementary estimates for linear stochastic equations. The first controls moments of a homogeneous linear equation and will be applied both to differences of flow trajectories and to the columns of the Jacobian matrix. The second treats an inhomogeneous equation and will be used to estimate spatial increments of the Jacobian. We state these estimates separately in order to keep the proof of the theorem focused on the spatial structure of the flow.

Lemma 2.2 (Moment estimate for linear stochastic equations)**.** *Let Y_t solve*

$$\begin{cases} dY_t = A_tY_t\,dt + \sum_{\ell=1}^r C_{\ell,t}Y_t\,dB_t^\ell,\\ Y_0 = y.\end{cases}$$

Assume that, for every $z\in\mathbb{R}^d$,

$$\langle z, A_t z\rangle \le L_x|z|^2, \qquad \sum_{\ell=1}^r |C_{\ell,t}z|^2 \le L_\sigma^2|z|^2.$$

Then, for every $q\ge 1$,

$$\mathbb{E}|Y_t|^{2q} \le |y|^{2q}\exp\left\{q\big(2L_x + (2q-1)L_\sigma^2\big)t\right\}.$$

Consequently,

$$\mathbb{E}\|J_t(u)\|_{\mathrm{op}}^{2q} \le d^q\exp\left\{q\big(2L_x + (2q-1)L_\sigma^2\big)t\right\}.$$

Proof. Applying Itô's formula to $|Y_t|^{2q}$, after a standard localization, gives

$$\begin{aligned}
d|Y_t|^{2q} &= 2q|Y_t|^{2q-2}\langle Y_t, A_tY_t\rangle\,dt\\
&\quad + q|Y_t|^{2q-2}\sum_{\ell=1}^r |C_{\ell,t}Y_t|^2\,dt\\
&\quad + 2q(q-1)|Y_t|^{2q-4}\sum_{\ell=1}^r \langle Y_t, C_{\ell,t}Y_t\rangle^2\,dt + dM_t,
\end{aligned}$$

where M_t is a local martingale. Since

$$\langle Y_t, C_{\ell,t} Y_t\rangle^2 \le |Y_t|^2 |C_{\ell,t} Y_t|^2,$$

the assumptions imply

$$d|Y_t|^{2q} \le q\big(2L_x + (2q-1)L_\sigma^2\big)|Y_t|^{2q}\,dt + dM_t.$$

Taking expectations in the localized equation and applying Gronwall's lemma yields

$$\mathbb{E}|Y_t|^{2q} \le |y|^{2q} \exp\left\{q\big(2L_x + (2q-1)L_\sigma^2\big)t\right\}.$$

For the matrix estimate, apply the vector estimate to $J_t(u)e_j$, $j = 1, \dots, d$, and use

$$\|J_t(u)\|_{\mathrm{op}}^{2q} \le \|J_t(u)\|_{\mathrm{HS}}^{2q} = \left(\sum_{j=1}^d |J_t(u)e_j|^2\right)^q \le d^{q-1}\sum_{j=1}^d |J_t(u)e_j|^{2q}.$$

This gives

$$\mathbb{E}\|J_t(u)\|_{\mathrm{op}}^{2q} \le d^q \exp\left\{q\big(2L_x + (2q-1)L_\sigma^2\big)t\right\}.$$

□

The preceding lemma applies directly to $X_t(u) - X_t(v)$, because the difference of two trajectories driven by the same Brownian motion satisfies a homogeneous linear equation with random coefficients. The difference $J_t(u) - J_t(v)$, however, satisfies an inhomogeneous linear equation: the additional terms arise from evaluating the derivative coefficients along the two different trajectories. We therefore record the corresponding moment estimate.

Lemma 2.3 (Inhomogeneous linear moment estimate)**.** *Let $m \ge 2$ be an even integer, and suppose that the matrix-valued process D_t satisfies*

$$dD_t = A_t D_t\,dt + \sum_{\ell=1}^r C_{\ell,t} D_t\,dB_t^\ell + F_t\,dt + \sum_{\ell=1}^r G_{\ell,t}\,dB_t^\ell.$$

Assume that, for every $M \in \mathbb{R}^{d\times d}$,

$$\langle M, A_t M\rangle_{\mathrm{HS}} \le L_x\|M\|_{\mathrm{HS}}^2, \qquad \sum_{\ell=1}^r \|C_{\ell,t}M\|_{\mathrm{HS}}^2 \le L_\sigma^2\|M\|_{\mathrm{HS}}^2.$$

Then there exist constants $C_m, c_m < \infty$, depending only on m, L_x, L_σ, such that

$$\begin{aligned}\mathbb{E}\|D_t\|_{\mathrm{HS}}^m &\le e^{c_m t}\mathbb{E}\|D_0\|_{\mathrm{HS}}^m \\ &\quad + C_m\int_0^t e^{c_m(t-s)}\left[\mathbb{E}\|F_s\|_{\mathrm{HS}}^m + \mathbb{E}\left(\sum_{\ell=1}^r \|G_{\ell,s}\|_{\mathrm{HS}}^2\right)^{m/2}\right] ds.\end{aligned}$$

Proof. Apply Itô's formula to $\|D_t\|_{\mathrm{HS}}^m$, after localization. Using

$$\sum_{\ell=1}^r \|C_{\ell,t}D_t + G_{\ell,t}\|_{\mathrm{HS}}^2 \le 2L_\sigma^2\|D_t\|_{\mathrm{HS}}^2 + 2\sum_{\ell=1}^r \|G_{\ell,t}\|_{\mathrm{HS}}^2$$

and Young's inequality gives

$$\begin{aligned}\frac{d}{dt}\mathbb{E}\|D_t\|_{\mathrm{HS}}^m &\le c_m\mathbb{E}\|D_t\|_{\mathrm{HS}}^m + C_m\mathbb{E}\|F_t\|_{\mathrm{HS}}^m \\ &\quad + C_m\mathbb{E}\left(\sum_{\ell=1}^r \|G_{\ell,t}\|_{\mathrm{HS}}^2\right)^{m/2}.\end{aligned}$$

The result follows from Gronwall's lemma. □

Although the drift depends on the random measure μ_t, differentiation with respect to the initial label acts only on the spatial argument $X_t(u)$. Because of that fact we can evaluate $\|J_t(u)\|$ using Kunita's theorem for the stochastic flows.

Under Assumptions 1 and 2, standard results on stochastic flows (see, e.g., [Kunita, 1986, Theorem 1.4.1]) yield a modification such that for a.s. ω and each $t \geq 0$ the map $u \mapsto T_t(\omega, u)$ is $\mathcal{C}^1$.

Proof of Theorem 2.2. Fix a compact set $K \subset \mathbb{R}^d$, $t \geq 0$, and $p \geq 1$. Choose an even integer $m > \max\{2p, 2d\}$. It is enough to prove (5) with p replaced by m, because then Hölder's inequality gives

$$\mathbb{E}\,\mathcal{K}_t(\cdot; K)^p \leq \big(\mathbb{E}\,\mathcal{K}_t(\cdot; K)^m\big)^{p/m}.$$

Fix $u, v \in \mathbb{R}^d$ and set

$$\Delta X_t := X_t(u) - X_t(v).$$

Since the same random measure μ_t appears in both equations, subtraction yields

$$d\Delta X_t = \big(a(X_t(u), \mu_t) - a(X_t(v), \mu_t)\big)\, dt + \sum_{\ell=1}^{r} \big(\sigma_\ell(X_t(u)) - \sigma_\ell(X_t(v))\big)\, dB_t^\ell.$$

Define

$$\widetilde{A}_t(u, v) := \int_0^1 D_x a\big(X_t(v) + \theta \Delta X_t, \mu_t\big)\, d\theta$$

and, for $\ell = 1, \dots, r$,

$$\widetilde{C}_{\ell,t}(u, v) := \int_0^1 D\sigma_\ell\big(X_t(v) + \theta \Delta X_t\big)\, d\theta.$$

By the fundamental theorem of calculus,

$$\begin{cases} d\Delta X_t = \widetilde{A}_t(u,v)\Delta X_t\, dt + \sum_{\ell=1}^{r} \widetilde{C}_{\ell,t}(u,v)\Delta X_t\, dB_t^\ell, \\ \Delta X_0 = u - v. \end{cases}$$

Moreover, by (2),

$$\|\widetilde{A}_t(u, v)\|_{\mathrm{op}} \leq L_x.$$

Jensen's inequality also gives, for every $z \in \mathbb{R}^d$,

$$\sum_{\ell=1}^{r} |\widetilde{C}_{\ell,t}(u, v)z|^2 \leq \int_0^1 \sum_{\ell=1}^{r} \big|D\sigma_\ell\big(X_t(v) + \theta \Delta X_t\big)z\big|^2\, d\theta \leq L_\sigma^2 |z|^2.$$

Applying Lemma 2.2 with $q = m/2$ and with $q = m$, respectively, yields

$$\mathbb{E}|X_t(u) - X_t(v)|^m \leq C_m e^{c_m t} |u - v|^m, \tag{11}$$

$$\mathbb{E}|X_t(u) - X_t(v)|^{2m} \leq C_m e^{c_m t} |u - v|^{2m}. \tag{12}$$

Lemma 2.2 gives, for every $q \geq 1$ and every $u \in \mathbb{R}^d$,

$$\mathbb{E}\,\|J_t(u)\|_{\mathrm{op}}^{2q} \leq d^q \exp\Big(q(2L + dL_\sigma^2)t + 2q^2 d L_\sigma^2 t\Big).$$

In particular, taking $q = m/2$ and $q = m$, we obtain

$$\mathbb{E}\,\|J_t(u)\|_{\mathrm{op}}^{m} \leq C_m e^{c_m t}, \qquad \mathbb{E}\,\|J_t(u)\|_{\mathrm{op}}^{2m} \leq C_m e^{c_m t}, \tag{13}$$

uniformly in $u \in \mathbb{R}^d$.

Step 2: Spatial increments of the Jacobian. Fix $u, v \in \mathbb{R}^d$, and write

$$D_t := J_t(u) - J_t(v).$$

Set

$$A_t(w) := D_x a(X_t(w), \mu_t), \qquad C_{\ell,t}(w) := D\sigma_\ell(X_t(w)), \qquad \ell = 1, \ldots, r.$$

Subtracting the Jacobian equations at u and v, we obtain

$$\begin{cases} dD_t = A_t(u) D_t \, dt + \sum_{\ell=1}^r C_{\ell,t}(u) D_t \, dB_t^\ell + F_t \, dt + \sum_{\ell=1}^r G_{\ell,t} \, dB_t^\ell, \\ D_0 = 0, \end{cases} \tag{14}$$

where

$$F_t := \big(A_t(u) - A_t(v)\big) J_t(v)$$

and

$$G_{\ell,t} := \big(C_{\ell,t}(u) - C_{\ell,t}(v)\big) J_t(v), \qquad \ell = 1, \ldots, r.$$

For every $M \in \mathbb{R}^{d \times d}$,

$$\langle M, A_t(u) M \rangle_{\mathrm{HS}} \le L_x \|M\|_{\mathrm{HS}}^2,$$

and

$$\sum_{\ell=1}^r \|C_{\ell,t}(u) M\|_{\mathrm{HS}}^2 \le L_\sigma^2 \|M\|_{\mathrm{HS}}^2.$$

Indeed, both inequalities follow by applying (2) to the columns of M.

Similarly, applying (4) columnwise gives

$$\|F_t\|_{\mathrm{HS}} \le H |\Delta X_t| \, \|J_t(v)\|_{\mathrm{HS}} \tag{15}$$

and

$$\left(\sum_{\ell=1}^r \|G_{\ell,t}\|_{\mathrm{HS}}^2 \right)^{1/2} \le H |\Delta X_t| \, \|J_t(v)\|_{\mathrm{HS}}.$$

Consequently, Hölder's inequality, (12), and (13) imply

$$\begin{aligned} &\mathbb{E}\|F_t\|_{\mathrm{HS}}^m + \mathbb{E} \left(\sum_{\ell=1}^r \|G_{\ell,t}\|_{\mathrm{HS}}^2 \right)^{m/2} \\ &\quad \le C_m \left(\mathbb{E}|\Delta X_t|^{2m} \right)^{1/2} \left(\mathbb{E}\|J_t(v)\|_{\mathrm{HS}}^{2m} \right)^{1/2} \\ &\quad \le C_m e^{c_m t} |u - v|^m. \end{aligned}$$

Here we also used $\|M\|_{\mathrm{HS}} \le \sqrt{d} \, \|M\|_{\mathrm{op}}$.

Applying Lemma 2.3 to (14), and using $D_0 = 0$, gives

$$\begin{aligned} \mathbb{E}\|D_t\|_{\mathrm{HS}}^m &\le C_m |u - v|^m \int_0^t e^{c_m (t-s)} e^{c_m s} \, ds \\ &\le C_m e^{c_m t} |u - v|^m, \end{aligned}$$

after enlarging c_m. Therefore,

$$\mathbb{E}\|J_t(u) - J_t(v)\|_{\mathrm{HS}}^m \le C_m e^{c_m t} |u - v|^m, \qquad u, v \in \mathbb{R}^d. \tag{16}$$

In particular, the same estimate holds with the operator norm in place of the Hilbert–Schmidt norm.

Step 3: Kolmogorov–Totoki and the compact supremum. Let $Q \subset \mathbb{R}^d$ be a compact cube containing K, and choose $u_0 \in K$. By (16), $u, v \in Q$

$$\mathbb{E}\|J_t(u) - J_t(v)\|_{\mathrm{HS}}^m \le C_m e^{c_m t}|u - v|^m.$$

Since $m > d$, this may be written as

$$\mathbb{E}\|J_t(u) - J_t(v)\|_{\mathrm{HS}}^m \le C_m e^{c_m t}|u - v|^{d+(m-d)}.$$

Hence the quantitative Kolmogorov–Totoki theorem, applied to the $\mathbb{R}^{d\times d}$-valued random field

$$u \longmapsto J_t(u) - J_t(u_0),$$

gives

$$\mathbb{E}\left[\sup_{u\in Q} \|J_t(u) - J_t(u_0)\|_{\mathrm{HS}}^m\right] \le C_{m,Q} e^{c_m t}. \tag{17}$$

Here we identify $\mathbb{R}^{d\times d}$ with $\mathbb{R}^{d^2}$.

Since Theorem 2.1 already provides a spatially continuous version of J_t, the continuous modification furnished by the Kolmogorov–Totoki theorem agrees with this version on Q almost surely.

For every $u \in K$,

$$\|J_t(u)\|_{\mathrm{op}} \le \|J_t(u_0)\|_{\mathrm{op}} + \|J_t(u) - J_t(u_0)\|_{\mathrm{HS}}.$$

Consequently,

$$\mathcal{K}_t(\cdot; K) \le \|J_t(u_0)\|_{\mathrm{op}} + \sup_{u\in Q} \|J_t(u) - J_t(u_0)\|_{\mathrm{HS}}.$$

Using (13), (17), and $(a + b)^m \le 2^{m-1}(a^m + b^m)$, we obtain

$$\mathbb{E}\mathcal{K}_t(\cdot; K)^m \le C_{m,K} e^{c_m t}.$$

Finally,

$$\mathbb{E}\mathcal{K}_t(\cdot; K)^p \le (\mathbb{E}\mathcal{K}_t(\cdot; K)^m)^{p/m} \le C_{p,K} e^{c_p t}.$$

This proves (5). □

3. The averaged Jacobian

The preceding pathwise estimates use the quantity $\mathcal{K}_t(\omega; K)$ which controls the worst possible stretching of the flow on the compact set K. Consider instead the averaged Jacobian for the transported measure $\mu_t = \mu_0 \circ T_t^{-1}$,

$$\mathcal{E}_t := \int_{\mathbb{R}^d} \|J_t(u)\|_{\mathrm{HS}}^2 \, \mu_0(du).$$

It measures average deformation of the flow over the initial mass, rather than the maximal deformation on the whole support.

Proposition 3.1 (Averaged Jacobian energy estimate)**.** *Assume that the flow is differentiable in the initial label and that there exist constants $L_{\mathrm{sym}}, \Gamma < \infty$ such that*

$$\langle z, \partial_x a(x,\mu) z\rangle \le L_{\mathrm{sym}} |z|^2, \qquad x, z \in \mathbb{R}^d, \ \mu \in \mathcal{P}_2(\mathbb{R}^d), \tag{18}$$

and

$$\sum_{k=1}^d |\partial_x \sigma^{(k)}(x) z|^2 \le \Gamma^2 |z|^2, \qquad x, z \in \mathbb{R}^d. \tag{19}$$

Then

$$\mathbb{E}\mathcal{E}_t \le d \exp\{(2L_{\mathrm{sym}} + \Gamma^2)t\}. \tag{20}$$

Proof. Recall that

$$dJ_t(u) = A_t(u) J_t(u)\, dt + \sum_{k=1}^d C_t^{(k)}(u) J_t(u)\, dB_t^k,$$

where

$$A_t(u) = \partial_x a(X_t(u), \mu_t), \qquad C_t^{(k)}(u) = \partial_x \sigma^{(k)}(X_t(u)).$$

Applying Itô's formula to $\|J_t(u)\|_{\mathrm{HS}}^2$ gives

$$\begin{aligned} d\|J_t(u)\|_{\mathrm{HS}}^2 = &\left[2\langle J_t(u), A_t(u) J_t(u)\rangle_{\mathrm{HS}} + \sum_{k=1}^d \|C_t^{(k)}(u) J_t(u)\|_{\mathrm{HS}}^2 \right] dt \\ &+ 2\sum_{k=1}^d \langle J_t(u), C_t^{(k)}(u) J_t(u)\rangle_{\mathrm{HS}}\, dB_t^k. \end{aligned}$$

Since

$$\langle J, A_t J\rangle_{\mathrm{HS}} = \sum_{i=1}^d \langle J e_i, A_t J e_i\rangle,$$

condition equation (18) gives

$$2\langle J, A_t J\rangle_{\mathrm{HS}} \le 2L_{\mathrm{sym}} \|J\|_{\mathrm{HS}}^2.$$

Similarly, equation (19) gives

$$\sum_{k=1}^d \|C_t^{(k)} J\|_{\mathrm{HS}}^2 = \sum_{i=1}^d \sum_{k=1}^d |C_t^{(k)} J e_i|^2 \le \Gamma^2 \|J\|_{\mathrm{HS}}^2.$$

Integrating with respect to $\mu_0(du)$ and taking expectations removes the martingale term, hence

$$\frac{d}{dt} \mathbb{E}\mathcal{E}_t \le (2L_{\mathrm{sym}} + \Gamma^2) \mathbb{E}\mathcal{E}_t.$$

Since $J_0(u) = I$, we have $\mathcal{E}_0 = \|I\|_{\mathrm{HS}}^2 = d$. Gronwall's lemma proves equation (20). □

To estimate $\mathbb{E} \sup_{u \in K} \|J_t(u)\|^p$, one must control increments $J_t(u) - J_t(v)$, which requires assumptions on the spatial regularity of $\partial_x a$ and $\partial \sigma$. The averaged quantity $\mathcal{E}_t$ does not compare two labels. It only estimates each $J_t(u)$ and integrates in u, so the one-sided bounds equations (18) and (19) are sufficient.

4. Internal fluctuations

This section concerns a separate kernel-type interacting model. In contrast to (1), both the drift and diffusion may depend on the order parameter Θ_t. Consequently, the assumptions of sections 2 and 3 are not imposed here.

$$\begin{cases} dX_t(u) &= \Big[b(X_t(u), \Theta_t) + \int_{\mathbb{R}^d} K(X_t(u), X_t(v), \Theta_t)\, \mu_0(dv)\Big] dt \\ &\quad + \sum_{\ell=1}^m \sigma_\ell(X_t(u), \Theta_t)\, dB_t^\ell, \\ X_0(u) &= u, \qquad u \in \mathbb{R}^d, \\ \mu_t &= \mu_0 \circ X_t^{-1}. \end{cases} \tag{21}$$

where

$$\theta(\mu) := \int_{\mathbb{R}^d} \Phi(x)\, \mu(dx),$$

Assumption 3 (Well-posedness of Equation (21))**.** *The maps b, K, σ_ℓ, Φ are globally Lipschitz and have at most linear growth. More explicitly, there exists $L < \infty$ such that*

$$\begin{aligned} |\Phi(x) - \Phi(y)| &\le L|x - y|, \\ |b(x, \theta) - b(y, \eta)| &\le L\big(|x - y| + |\theta - \eta|\big), \\ |K(x, z, \theta) - K(y, w, \eta)| &\le L\big(|x - y| + |z - w| + |\theta - \eta|\big), \end{aligned}$$

and

$$\sum_{\ell=1}^m |\sigma_\ell(x, \theta) - \sigma_\ell(y, \eta)|^2 \le L^2\big(|x - y|^2 + |\theta - \eta|^2\big).$$

Under Assumption 3, equation (21) admits a unique global continuous solution.

This section studies the pairwise energy

$$\mathcal{D}_t := \iint_{\mathbb{R}^d \times \mathbb{R}^d} |X_t(u) - X_t(v)|^2\, \mu_0(du)\mu_0(dv).$$

4.1. Dissipativity assumptions.

Equivalently, if

$$\bar{X}_t := \int_{\mathbb{R}^d} X_t(u)\, \mu_0(du),$$

and

$$\mathcal{V}_t := \int_{\mathbb{R}^d} |X_t(u) - \bar{X}_t|^2\, \mu_0(du),$$

then

$$\mathcal{D}_t = 2\mathcal{V}_t.$$

Indeed,

$$\begin{aligned} \mathcal{D}_t &= \iint |X_t(u) - X_t(v)|^2\, \mu_0(du)\mu_0(dv) \\ &= \iint \big(|X_t(u)|^2 + |X_t(v)|^2 - 2\langle X_t(u), X_t(v)\rangle\big)\, \mu_0(du)\mu_0(dv) \\ &= 2\int |X_t(u)|^2\, \mu_0(du) - 2\left|\int X_t(u)\, \mu_0(du)\right|^2 \\ &= 2\int |X_t(u) - \bar{X}_t|^2\, \mu_0(du) = 2\mathcal{V}_t. \end{aligned}$$

We assume the following dissipativity condition.

Assumption 4 (Pairwise dissipativity). *There exists $\lambda > 0$ such that for every $x, y \in \mathbb{R}^d$, every $\theta \in \mathbb{R}^q$, and every $\nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$2\left\langle x-y, b(x,\theta)-b(y,\theta)+\int_{\mathbb{R}^d}\big[K(x,z,\theta)-K(y,z,\theta)\big]\nu(dz)\right\rangle + \sum_{\ell=1}^{m}|\sigma_\ell(x,\theta)-\sigma_\ell(y,\theta)|^2 \le -\lambda|x-y|^2.$$

The dissipativity condition covers several basic models.

Example 4.1 (Ornstein–Uhlenbeck SDEWI). Consider

$$dX_t(u) = -\lambda_0\big(X_t(u)-m_t\big)\,dt + \sigma(X_t(u))\,dB_t, \qquad m_t = \int_{\mathbb{R}^d} X_t(v)\,\mu_0(dv).$$

If

$$\|\sigma(x)-\sigma(y)\|_{\mathrm{HS}} \le L_\sigma|x-y|,$$

then

$$\mathbb{E}\mathcal{D}_t \le e^{-(2\lambda_0-L_\sigma^2)t}\mathcal{D}_0.$$

Thus the cloud contracts whenever $2\lambda_0 > L_\sigma^2$.

Example 4.2 (Gradient systems). Take

$$b(x,\theta) = -\nabla V(x), \qquad K(x,y,\theta) = -\nabla W(x-y).$$

Then

$$dX_t(u) = \left[-\nabla V(X_t(u)) - \int_{\mathbb{R}^d}\nabla W(X_t(u)-X_t(v))\,\mu_0(dv)\right]dt + \sigma(X_t(u))\,dB_t.$$

If

$$D^2V \ge \rho I, \qquad D^2W \ge \kappa I, \qquad \|\sigma(x)-\sigma(y)\| \le L_\sigma|x-y|,$$

then

$$\mathbb{E}\mathcal{D}_t \le e^{-(2\rho+2\kappa-L_\sigma^2)t}\mathcal{D}_0.$$

Example 4.3 (Nonlinear statistics). Let

$$\Theta_t = \int_{\mathbb{R}^d}\psi(X_t(v))\,\mu_0(dv),$$

and consider

$$dX_t(u) = -\lambda_0\big(X_t(u)-\Theta_t\big)\,dt + \sigma(X_t(u),\Theta_t)\,dB_t.$$

For two labels u, v, the order parameter cancels in the drift difference:

$$-\lambda_0\big(X_t(u)-\Theta_t\big) + \lambda_0\big(X_t(v)-\Theta_t\big) = -\lambda_0\big(X_t(u)-X_t(v)\big).$$

4.2. **Proof.**

Theorem 4.1 (Synchronization under dissipativity assumptions). *Assume that $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$ and that Assumption 4 holds. Then*

$$\mathbb{E}\mathcal{D}_t \le e^{-\lambda t}\mathcal{D}_0.$$

In particular,

$$\mathbb{E}W_2^2(\mu_t,\delta_{\bar{X}_t}) \le e^{-\lambda t}W_2^2(\mu_0,\delta_{\bar{X}_0}).$$

Thus the transported measure μ_t concentrates around its random center of mass $\bar{X}_t$.

Proof. Fix two labels $u, v \in \mathbb{R}^d$. Define the difference

$$Z_t^{u,v} := X_t(u) - X_t(v).$$

We first prove that under Assumption 4

$$\mathbb{E}|Z_t^{u,v}|^2 \le e^{-\lambda t}|u-v|^2,$$

then we integrate this estimate with respect to $\mu_0(du)\mu_0(dv)$.

Writing equation (21) for $X_t(u)$ and for $X_t(v)$ and then subtracting the two equations gives

$$dZ_t^{u,v} = G_t^{u,v}\,dt + \sum_{\ell=1}^m H_{\ell,t}^{u,v}\,dB_t^\ell,$$

where the drift coefficient

$$\begin{aligned}G_t^{u,v} &:= b(X_t(u),\Theta_t) - b(X_t(v),\Theta_t)\\ &\quad + \int_{\mathbb{R}^d}\big[K(X_t(u),X_t(w),\Theta_t) - K(X_t(v),X_t(w),\Theta_t)\big]\,\mu_0(dw),\end{aligned}$$

and the diffusion coefficient is

$$H_{\ell,t}^{u,v} := \sigma_\ell(X_t(u),\Theta_t) - \sigma_\ell(X_t(v),\Theta_t).$$

Since $\mu_t = \mu_0 \circ X_t^{-1}$, we may rewrite the interaction term as

$$\int_{\mathbb{R}^d} K(X_t(u),X_t(w),\Theta_t)\,\mu_0(dw) = \int_{\mathbb{R}^d} K(X_t(u),z,\Theta_t)\,\mu_t(dz).$$

Thus

$$\begin{aligned}G_t^{u,v} &:= b(X_t(u),\Theta_t) - b(X_t(v),\Theta_t)\\ &\quad + \int_{\mathbb{R}^d}\big[K(X_t(u),z,\Theta_t) - K(X_t(v),z,\Theta_t)\big]\,\mu_t(dz).\end{aligned}$$

Applying Itô's formula to $|Z_t^{u,v}|^2$, we obtain

$$d|Z_t^{u,v}|^2 = 2\langle Z_t^{u,v}, G_t^{u,v}\rangle\,dt + \sum_{\ell=1}^m |H_{\ell,t}^{u,v}|^2\,dt + 2\sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v}\rangle\,dB_t^\ell.$$

Apply Assumption 4 with $x = X_t(u)$, $y = X_t(v)$, $\theta = \Theta_t$, and $\nu = \mu_t$. Then

$$2\langle Z_t^{u,v}, G_t^{u,v}\rangle + \sum_{\ell=1}^m |H_{\ell,t}^{u,v}|^2 \le -\lambda|Z_t^{u,v}|^2.$$

Therefore

$$d|Z_t^{u,v}|^2 \le -\lambda|Z_t^{u,v}|^2\,dt + 2\sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v}\rangle\,dB_t^\ell. \tag{22}$$

Set the new variable

$$Y_t^{u,v} := e^{\lambda t}|Z_t^{u,v}|^2.$$

By Itô's formula,

$$\begin{aligned}dY_t^{u,v} &= e^{\lambda t}\left[\lambda|Z_t^{u,v}|^2 + 2\langle Z_t^{u,v}, G_t^{u,v}\rangle + \sum_{\ell=1}^m |H_{\ell,t}^{u,v}|^2\right]dt\\ &\quad + 2e^{\lambda t}\sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v}\rangle\,dB_t^\ell.\end{aligned}$$

By the drift bound equation (22), the drift term above is nonpositive. Hence

$$dY_t^{u,v} \le 2e^{\lambda t}\sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v}\rangle\,dB_t^\ell.$$

To make this completely rigorous, introduce the localization sequence

$$\tau_R := \inf\{t \geq 0 : |X_t(u)| + |X_t(v)| + |\Theta_t| \geq R\}.$$

Since the solution is global and continuous, $\tau_R \uparrow \infty$ almost surely as $R \to \infty$. Stopping this inequality at τ_R, we get

$$Y^{u,v}_{t\wedge\tau_R} \leq Y^{u,v}_0 + 2\sum_{\ell=1}^{m} \int_0^{t\wedge\tau_R} e^{\lambda s}\langle Z^{u,v}_s, H^{u,v}_{\ell,s}\rangle\, dB^\ell_s.$$

The stopped stochastic integral is a true martingale. Therefore, taking expectations gives

$$\mathbb{E}Y^{u,v}_{t\wedge\tau_R} \leq Y^{u,v}_0.$$

Since $Y^{u,v}_0 = |u-v|^2$, we have

$$\mathbb{E}\left[e^{\lambda(t\wedge\tau_R)}|Z^{u,v}_{t\wedge\tau_R}|^2\right] \leq |u-v|^2.$$

Letting $R \to \infty$ and using Fatou's lemma yields

$$\mathbb{E}\left[e^{\lambda t}|Z^{u,v}_t|^2\right] \leq |u-v|^2.$$

Therefore

$$\mathbb{E}|X_t(u) - X_t(v)|^2 \leq e^{-\lambda t}|u-v|^2.$$

This proves the pairwise synchronization estimate.

Now integrate the pairwise synchronization estimate with respect to $\mu_0(du)\mu_0(dv)$. Since $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathcal{D}_0 = \iint |u-v|^2\, \mu_0(du)\mu_0(dv) < \infty.$$

By Fubini's theorem,

$$\begin{aligned}
\mathbb{E}\mathcal{D}_t &= \mathbb{E}\iint |X_t(u) - X_t(v)|^2\, \mu_0(du)\mu_0(dv) \\
&= \iint \mathbb{E}|X_t(u) - X_t(v)|^2\, \mu_0(du)\mu_0(dv) \\
&\leq e^{-\lambda t}\iint |u-v|^2\, \mu_0(du)\mu_0(dv).
\end{aligned}$$

Hence $\mathbb{E}\mathcal{D}_t \leq e^{-\lambda t}\mathcal{D}_0$, proving the first estimate in the theorem.

Therefore

$$\mathbb{E}\mathcal{V}_t = \frac{1}{2}\mathbb{E}\mathcal{D}_t \leq \frac{1}{2}e^{-\lambda t}\mathcal{D}_0 = e^{-\lambda t}\mathcal{V}_0.$$

This proves the internal fluctuation decay estimate.

Finally, since $\mu_t = \mu_0 \circ X_t^{-1}$,

$$\int_{\mathbb{R}^d} |x - \bar{X}_t|^2\, \mu_t(dx) = \int_{\mathbb{R}^d} |X_t(u) - \bar{X}_t|^2\, \mu_0(du) = \mathcal{V}_t.$$

For the Dirac measure $\delta_{\bar{X}_t}$,

$$W_2^2(\mu_t, \delta_{\bar{X}_t}) = \int_{\mathbb{R}^d} |x - \bar{X}_t|^2\, \mu_t(dx).$$

Therefore

$$W_2^2(\mu_t, \delta_{\bar{X}_t}) = \mathcal{V}_t.$$

Taking expectations and using the previous estimate gives

$$\mathbb{E}W_2^2(\mu_t, \delta_{\bar{X}_t}) \leq e^{-\lambda t}W_2^2(\mu_0, \delta_{\bar{X}_0}).$$

□

This result is the contractive counterpart of the distortion estimates in the preceding sections. Sections 2 and 3 control the growth of infinitesimal perturbations, either uniformly or on average. Here the pairwise dissipativity assumption controls finite separations directly and forces their average to decay. Since $\mathcal{D}_t = 2\mathcal{V}_t$, this finite-scale contraction is exactly the collapse of the transported measure around its random center of mass.

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